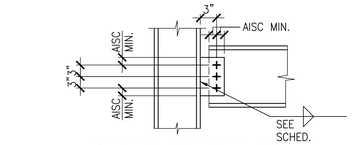
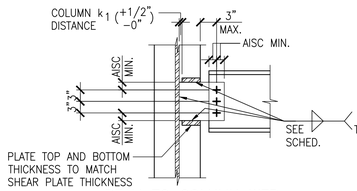
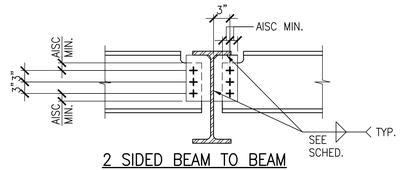


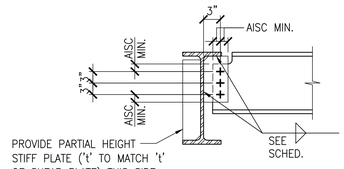
BEAM TO COLUMN FLANGE



BEAM TO COLUMN WEB



2 SIDED BEAM TO BEAM

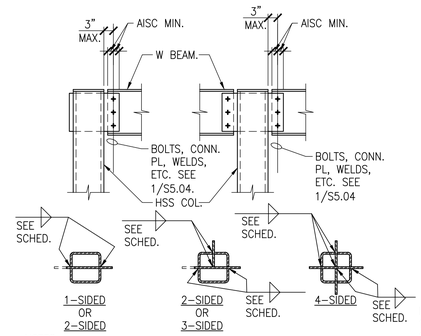


1 SIDED BEAM TO BEAM

BEAM CONNECTION SCHEDULE–ROOF				
NOMINAL MEMBER DEPTH	BOLTS NO. AND SIZE	SHEAR PLATE THICKNESS	SIZE OF FILLET WELD	REMARKS
8" – 10"	2–3/4" DIA.	3/8	5/16	–
12" – 14"	3–3/4" DIA.	3/8	5/16	–
16"	4–3/4" DIA.	3/8	5/16	–
18"	5–3/4" DIA.	3/8	5/16	–
21"	6–3/4" DIA.	3/8	5/16	–
24"	7–3/4" DIA.	3/8	5/16	–
27"	8–3/4" DIA.	3/8	5/16	–
30" – 33"	9–1" DIA.	9/16	7/16	–
36" – 40"	10–1" DIA.	9/16	7/16	–

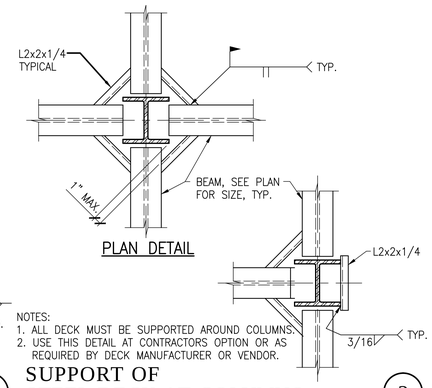
NOTES:
1. ALL 3/4" DIA. BOLTS SHALL BE A325–N. ALL 1" DIA. BOLTS SHALL BE A490–N.
2. PROVIDE LARGER WELDS WHERE MAY BE REQUIRED BY AISC.
3. USE LARGER PLATES AND WELDS WHERE REQUIRED BY BRACE FRAME CONNECTIONS OR OTHER SPECIFIC DETAILS.

TYPICAL BEAM CONNECTION AND DETAILS (1)
NTS



NOTES:
1. DRILL STRESS RELIEVING HOLE PRIOR TO SLOTTING TUBE, HOLE SHALL BE 1/8" LARGER THAN PLATE WIDTH.

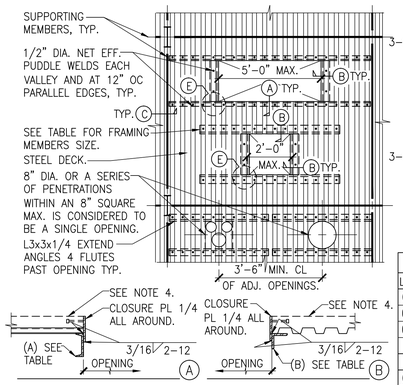
W BEAM TO TUBE COLUMN (2)
NTS

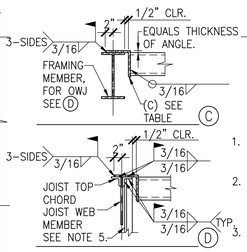


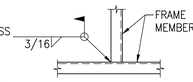
PLAN DETAIL

NOTES:
1. ALL DECK MUST BE SUPPORTED AROUND COLUMNS.
2. USE THIS DETAIL AT CONTRACTORS OPTION OR AS REQUIRED BY DECK MANUFACTURER OR VENDOR.

SUPPORT OF STEEL DECK AT COLUMNS (3)
NTS





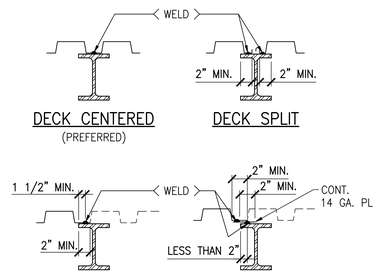


* TYPICAL ANGLE TO ANGLE OR ANGLE TO CHANNEL WELD. (E)

FRAME MEMBER SIZES.	
LONG OPENINGS DIM. > 24" < 60" MAX.	
(A)	C6x8.2
(B)	L4x3x1/4 LLV
(C)	L7x4x1/2x0'–3" LLV MIN.
LONG OPENINGS DIM. 24" MAX.	
(A)	L3x3x1/4

NOTES:
1. ADDED FRAMING IS NOT REQD. FOR OPENINGS WHERE LONG OPENING DIM. IS LESS THAN 6".
2. FOR STRUCTURE REQUIRED TO SUPPORT MECHANICAL UNIT GRAVITY AND LATERAL LOADS AND CURBS ETC. SEE MECH. AND ARCH. DRAWINGS.
3. SEE MECH. FOR EQUIPMENT MOUNTING, ATTACHMENT OPENINGS, ETC.
4. COMPOSITE SLAB OR ROOFING SEE PLANS AND ARCH. DRAWINGS. WHERE HARD ROCK CONC. OCCURS PROVIDE 3/4" DIA. X 8" NELSON HEADED STUDS AT 24" OC.
5. CONNECTION AT TOP CHORD PANEL POINT. IF NOT AT PANEL POINT, FIELD INSTALL L2X2X1/4 FROM POINT OF LOAD TO NEAREST PANEL POINT ON BOTTOM CHORD. SHORE JOIST DURING WELDING.

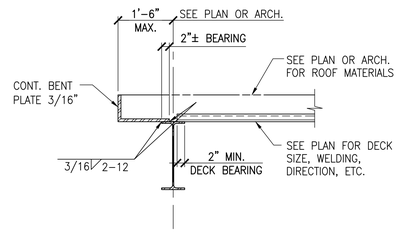
TYPICAL OPENIING IN STEEL DECK (4)
NTS



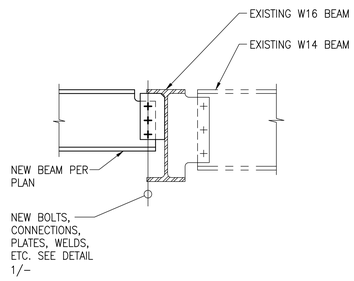
DECK NOT CENTERED
WELDS AT DECK PARALLEL TO BEAMS

NOTES:
1. SEE STEEL DECK WELDING SCHEDULE, NOTES AND OTHER DETAILS.

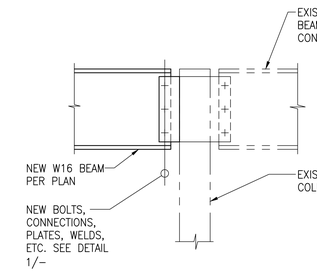
DECK WELDING DETAIL (5)
NTS



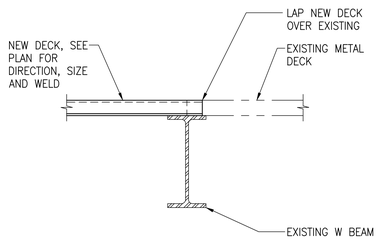
EDGE OF ROOF DECK (6)
NTS



BEAM TO EXISTING BEAM (7)
NTS

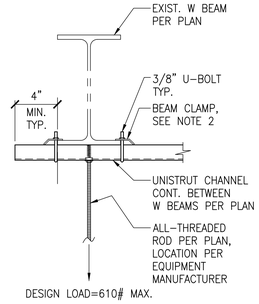


BEAM TO EXISTING COLUMN (8)
NTS



NOTE:
1. REMOVE EXISTING PARAPET AND ROOF EDGE BENT PLATE ALONG EXISTING BEAM AS REQUIRED.

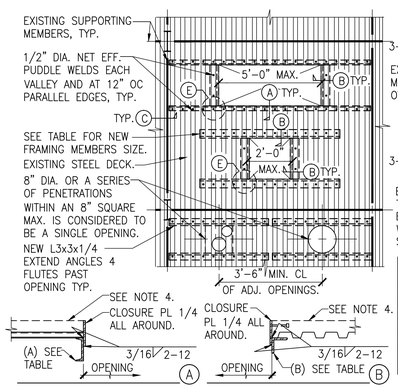
NEW STEEL DECK TO EXISTING STEEL DECK (9)
NTS

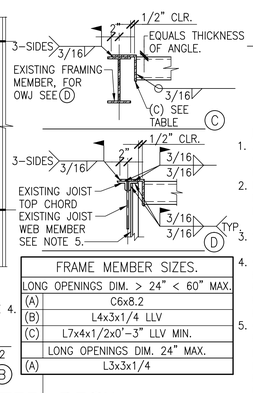


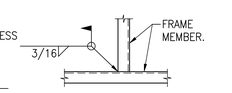
DESIGN LOAD=610# MAX.

NOTES:
1. REMOVE FIRE PROOFING FROM EXISTING BEAM AS REQUIRED.
2. USE P2785 CLAMP WITH P1000 CHANNEL AND P2786 CLAMP WITH P5500 CHANNEL.

HANGER ATTACHMENT TO EXISTING W BEAM (10)
NTS





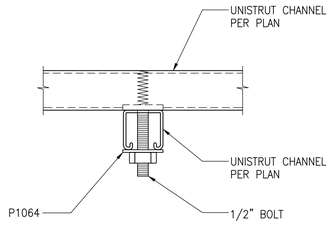


* TYPICAL ANGLE TO ANGLE OR ANGLE TO CHANNEL WELD. (E)

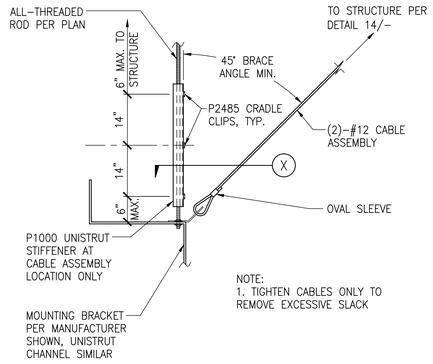
FRAME MEMBER SIZES.	
LONG OPENINGS DIM. > 24" < 60" MAX.	
(A)	C6x8.2
(B)	L4x3x1/4 LLV
(C)	L7x4x1/2x0'–3" LLV MIN.
LONG OPENINGS DIM. 24" MAX.	
(A)	L3x3x1/4

NOTES:
1. ADDED FRAMING IS NOT REQD. FOR OPENINGS WHERE LONG OPENING DIM. IS LESS THAN 6".
2. FOR STRUCTURE REQUIRED TO SUPPORT MECHANICAL UNIT GRAVITY AND LATERAL LOADS AND CURBS ETC. SEE MECH. AND ARCH. DRAWINGS.
3. SEE MECH. FOR EQUIPMENT MOUNTING, ATTACHMENT OPENINGS, ETC.
4. COMPOSITE SLAB OR ROOFING SEE PLANS AND ARCH. DRAWINGS. WHERE HARD ROCK CONC. OCCURS. SAW CUT PROPERLY AND PROVIDE CLOSURE PLATE AS SHOWN.
5. CONNECTION AT TOP CHORD PANEL POINT. IF NOT AT PANEL POINT, FIELD INSTALL L2X2X1/4 FROM POINT OF LOAD TO NEAREST PANEL POINT ON BOTTOM CHORD. SHORE JOIST DURING WELDING.

TYPICAL OPENING IN EXISTING STEEL DECK (11)
NTS

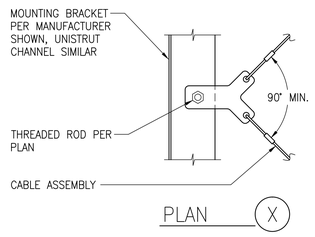


UNISTRUT TO UNISTRUT CONNECTION (12)
NTS

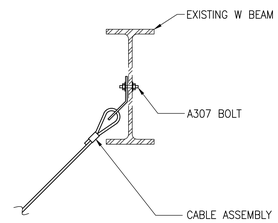


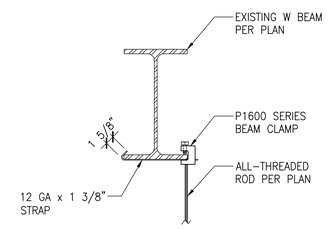
NOTE:
1. TIGHTEN CABLES ONLY TO REMOVE EXCESSIVE SLACK

SEISMIC BRACING DETAIL (13)
NTS



VENTILATION HOOD SEISMIC BRACING DETAIL (14)
NTS





HANGER ATTACHMENT TO EXISTING W BEAM (15)
NTS

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1	REVISIONS 06/06/11
2	PERMIT REVIEW 06/27/11
3	PERMIT REVIEW 07/08/11

CONSTRUCTION DOCUMENTS

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KEY:

SHEET TITLE: STEEL DETAILS

SHEET NUMBER: S5.02

DRAWN BY: EK	REVIEWED BY: SN/MT
DATE: 06/06/11	PROJECT NUMBER: 1101.00